Shreyansh Shekhar Dwivedi

Backend Developer | Java & Spring Boot

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PROFESSIONAL SUMMARY

B.Tech Computer Science (Data Science) student with strong expertise in backend engineering, AI-powered systems, and secure distributed architectures. Proven open-source contributor to large-scale repositories including TensorFlow (Google) and Eclipse Foundation. Experienced in building high-reliability RESTful APIs, implementing Retrieval-Augmented Generation (RAG) pipelines, and integrating cloud-native DevOps workflows using Docker, Kubernetes, and CI/CD pipelines. Passionate about developing scalable, production-grade software solutions.

TECHNICAL SKILLS

Programming Languages: Java (Advanced), Rust (Proficient), Bash/Shell Scripting

Backend & AI: Spring Boot, Spring AI, RESTful APIs, RAG Architecture, WebSockets (STOMP), OOP, Data Structures & Algorithms

DevOps & Cloud: Docker, Kubernetes, Linux, Maven, Git, CI/CD Pipelines

Databases: PostgreSQL, MongoDB

Security & Cryptography: AES-256 GCM, SHA-256, Quantum Key Distribution (BB84 Simulation), Penetration Testing Fundamentals

AI & LLM Integration: Retrieval-Augmented Generation (RAG), Embedding-based Similarity Search, OpenAI API, Gemini API

PROJECTS

Multi-Modal RAG & AI Chat Service — *Jan 2026*

- Built a Retrieval-Augmented Generation (RAG) pipeline enabling context-aware querying over private datasets using embedding-based similarity search, significantly reducing LLM hallucinations.
- Engineered custom Spring @Configuration with @Qualifier annotations to manage multi-provider AI beans (Gemini & OpenAI), enabling modular and scalable AI integrations.
- Implemented secure credential management using environment variables ensuring production-ready deployment best practices.
- Designed RESTful API backend architecture supporting multiple LLM providers and scalable document retrieval.

Tech Stack: *Java 21, Spring Boot, Spring AI, REST, RAG, OpenAI API, Gemini API, Embedding Search*

Hybrid Quantum Key Distribution (QKD) Simulator API — *Nov 2025*

- Designed and implemented a RESTful API simulating the BB84 protocol for secure quantum key exchange, including QBER (Quantum Bit Error Rate) validation and error correction mechanisms.
- Derived ephemeral AES-256 GCM session keys from the quantum key exchange process for quantum-resistant symmetric encryption.
- Demonstrated end-to-end secure key lifecycle management including key generation, distribution, error correction, and cryptographic protocol implementation.

Tech Stack: *Java 21, Spring Boot, REST, AES-256 GCM, SHA-256, BB84 Protocol*

OPEN SOURCE EXPERIENCE

Open Source Contributor — TensorFlow (Google Open Source)

- Resolved CI/CD 'run format check' pipeline failures, improving build reliability for a global multi-thousand contributor repository.
- Automated code-style enforcement by integrating Maven Spotless plugin, ensuring consistent formatting standards across the large-scale codebase.
- Contributed to maintaining code quality and consistency in one of the world's most widely used machine learning frameworks.

Open Source Contributor — Eclipse Foundation

- Improved project infrastructure and developer tooling to enhance contributor onboarding and productivity.
- Refactored test suites to improve maintainability, code coverage, and stability across multiple environments.
- Resolved configuration issues to ensure seamless cross-environment integration and compatibility.

EDUCATION

B.Tech — Computer Science Engineering (Data Science) — *Expected Graduation: 2028*

JSS Academy of Technical Education, Noida, Uttar Pradesh, India

ACHIEVEMENTS & HIGHLIGHTS

- Contributed to enterprise-scale open-source repositories including TensorFlow (Google) and Eclipse Foundation.
- Built secure distributed backend systems applying advanced cryptography (AES-256 GCM, SHA-256, QKD simulation).
- Designed AI-driven backend architectures integrating multiple LLM providers (OpenAI, Gemini) using RAG pipelines.
- Strong foundation in system design, clean code principles, and scalable microservices architecture.
- Actively solving Data Structures & Algorithms problems; strong interest in Cloud Native infrastructure and distributed systems.